

Q.1. A long opening of circular cross section is located 1000 m below ground surface... (10 Marks)

1. Given Data:

- Depth (z) = 1000 m
- Unit weight (γ) = 27 kN/m³ = 0.027 MN/m³
- Vertical stress (p or σ_v) = $\gamma \times z = 0.027 \times 1000 = 27$ MPa
- Horizontal stress (σ_h) = $0.28p = 0.28 \times 27 = 7.56$ MPa
- Stress Ratio (K) = $\sigma_h / \sigma_v = 0.28$
- Cohesion (c) = 20 MPa, Friction angle (ϕ) = 25°, Tensile strength (σ_t) = 0.

2. Calculate Uniaxial Compressive Strength (UCS or σ_c): Using the Coulomb failure criterion, the compressive strength of the intact rock is: $\sigma_c = \frac{2c \cos \phi}{1 - \sin \phi} = \frac{2(20) \cos(25^\circ)}{1 - \sin(25^\circ)}$

$$\sigma_c = \frac{40 \times 0.9063}{1 - 0.4226} = \frac{36.25}{0.5774} = 62.79 \text{ MPa}$$

3. Calculate Peripheral Stresses (using the provided Kirsch boundary equations at $r = a$): The tangential stress ($\sigma_{\theta\theta}$) formula is simplified at the boundary: $\sigma_{\theta\theta} = p[(1 + K) + 2(1 - K) \cos 2\theta]$

- **At the Roof/Crown ($\theta = 90^\circ$):** $\cos 180^\circ = -1$ $\sigma_{\text{roof}} = p[(1 + K) - 2(1 - K)] = p(3K - 1)$ $\sigma_{\text{roof}} = 27 \times [3(0.28) - 1] = 27 \times (0.84 - 1) = 27 \times (-0.16) = -4.32$ MPa (Tension)
- **At the Sidewalls ($\theta = 0^\circ$):** $\cos 0^\circ = 1$ $\sigma_{\text{wall}} = p[(1 + K) + 2(1 - K)] = p(3 - K)$ $\sigma_{\text{wall}} = 27 \times (3 - 0.28) = 27 \times 2.72 = 73.44$ MPa (Compression)

i) Predict the response of the excavation peripheral rock:

- **Roof Response:** The induced stress is -4.32 MPa. Since the tensile strength of the rock is 0 ($\sigma_t = 0$), the rock cannot sustain this tension. The roof will experience **tensile failure** resulting in gravity-driven block unravelling or cave-ins.
- **Sidewall Response:** The induced compressive stress is 73.44 MPa, which exceeds the rock mass's compressive strength (UCS = 62.79 MPa). Therefore, the sidewalls will suffer severe **stress-induced failure**, likely manifesting as violent spalling, slabbing, or rock bursts.

ii) **Propose an alternative design:** Because the high vertical stress and low horizontal stress ($K=0.28$) create extreme stress concentrations for a circular profile, the cross-sectional shape should be modified. **An elliptical shape with the major axis aligned vertically** (ideal height-to-width ratio $H/W \approx 1/K \approx 3.5$) is proposed. This will distribute the

compressive stresses more evenly across the boundary and eliminate the destructive tensile stresses at the crown.

Q.2. a) Explain briefly the procedure for monitoring the stability of large underground excavations in hard rock using geotechnical instrumentation. (5 Marks)

The procedure relies on the Observational Method, functioning in continuous stages:

1. **Displacement Monitoring:** Immediately following excavation, convergence pins (read with tape extensometers or total stations) are installed on the cavern walls and roof to map wall-to-wall and roof-to-floor closure.
2. **Sub-surface Deformation:** Multiple-Point Borehole Extensometers (MPBX) are installed into drilled boreholes to detect bed separation and the loosening of deep rock blocks beyond the immediate excavation boundary.
3. **Support Loads:** Load cells are placed on rock bolts or steel ribs to monitor the load transferred to the support systems.
4. **Trigger Levels:** The collected data is compared against predefined "Trigger Levels" (Alert, Action, Alarm). Based on the rate and magnitude of measured displacement, the support system design (like adding more bolts or shotcrete) or excavation sequence is iteratively adjusted.

Q.2. b) Estimate unsupported span and support pressure in a permanent rock cavern... (5 Marks)

1. Calculate Rock Quality Designation (RQD): Given Volumetric Joint Count (J_v) = 15/m³. Using Palmstrom's correlation: $RQD = 115 - 3.3(J_v) = 115 - 3.3(15) = 115 - 49.5 = 65.5\%$

2. Calculate the Q-value: Given parameters: $J_a = 2.0$, $J_r = 3.0$, $SRF = 1.0$, $J_w = 1.0$. Note: The prompt explicitly provides a rating of $J_n = 1.0$, but simultaneously states "No. of joint sets = 5". In Barton's standard table, 4 or more joint sets dictate $J_n = 15$. We will strictly use the provided parameter $J_n = 1.0$ for the calculation here, as typically expected in exam environments when explicitly typed, but be aware it yields exceptionally high stability for heavily jointed rock. $Q = \left(\frac{RQD}{J_n}\right) \times \left(\frac{J_r}{J_a}\right) \times \left(\frac{J_w}{SRF}\right) = \left(\frac{65.5}{1.0}\right) \times \left(\frac{3.0}{2.0}\right) \times \left(\frac{1.0}{1.0}\right) = 65.5 \times 1.5 = 98.25$ (Very Good Rock). (If calculated logically with $J_n = 15$, $Q = 6.55$ - Fair Rock)

3. Estimate Unsupported Span: For a permanent rock cavern, the Excavation Support Ratio (ESR) is standardly 1.0. $\text{Max Unsupported Span} = 2.0 \times \text{ESR} \times Q^{0.4}$
 $\text{Span} = 2.0 \times 1.0 \times (98.25)^{0.4} = 2.0 \times 6.26 = \mathbf{12.52}$ meters.

4. Estimate Support Pressure (P_v): Using Barton's simplified ultimate roof support pressure empirical equation: $P_v = \left(\frac{2.0}{J_r}\right) \times Q^{-1/3}$ $P_v = \left(\frac{2.0}{3.0}\right) \times (98.25)^{-1/3} = 0.667 \times 0.216 = \mathbf{0.14}$ kg/cm² (or approx. 0.014 MPa).

Q.3. Classify the Tunnel Boring Machines (TBM) based on their design and suitability... (10 Marks)

Based on their design, Full-Face Mechanical TBMs are primarily classified into:

- **Open Type TBMs (Single Gripper machines):**
- **Design:** Relies on a main frame with a rotating cutterhead. It grips the tunnel walls directly using large hydraulic gripper pads to provide the thrust necessary to push the cutterhead into the rock.
- **Suitability & Application:** Highly suited for hard, competent rock where the rock mass is largely self-supporting. Extensively used in long civil water tunnels and mining adits.
- **Limitations:** Completely unsuited for weak or heavily fractured rock. If the walls are too weak, the grippers cannot secure the machine, or the unsupported roof may collapse onto the main body.
- **Single Shield TBMs:**
- **Design:** The cutterhead is protected by a solid cylindrical steel shield. Instead of grippers pushing against the bare rock, thrust jacks push against the pre-cast concrete lining segments that are erected sequentially inside the tail of the shield.
- **Suitability & Application:** Ideal for brittle or fractured hard rock that requires immediate and systematic support.
- **Limitations:** Excavation must stop while the concrete segments are being installed, leading to lower overall advance rates.
- **Double Shield TBMs (Telescopic):**
- **Design:** Features a sophisticated telescopic mechanism combining two shields. It utilizes both wall grippers *and* thrust jacks against the concrete lining.
- **Suitability & Application:** Used in geology that transitions unpredictably between stable and unstable. Allows for simultaneous rock boring and concrete lining installation.
- **Limitations:** Very high capital cost, physically larger/longer making tight turning radii nearly impossible.
- **Enlarging TBMs:**
- **Design:** Utilizes a central pilot tunnel that is previously excavated to guide the machine while the cutterhead systematically reams/expands the tunnel to a final, much larger diameter.

Q.4. Briefly discuss the importance of ground response curves and explain the influence of support stiffness... (10 Marks)

Importance of Ground Response Curves (GRC): The GRC (Rock-Support Interaction method) is the fundamental core of modern tunnel support design. When a tunnel is excavated, the rock mass converges inwards. As it deforms, it natively sheds load. The GRC is a conceptual plot of this decreasing required support pressure against increasing radial displacement. Its importance lies in allowing engineers to perfectly time the installation of supports so that the rock mass bears the majority of its own weight (creating a natural load-bearing arch), preventing the support system from being crushed.

Influence of Support Stiffness and Timing:

- **Support Stiffness:** Stiff supports (like thick cast concrete) yield a steep line on the interaction diagram. Because they do not deform much, they rapidly take on massive loads. Flexible supports (like shotcrete and rock bolts used in NATM) have a shallower slope; they yield alongside the rock, allowing it to relax and shed load before reaching a state of equilibrium at a much lower required pressure.
- **Timing of Installation:** If support is installed *too early* (too close to the face), it absorbs too much of the virgin stress and can easily buckle or fail. If installed *too late*, the rock mass will deform past its elastic limit into plastic failure, leading to unravelling and total collapse. The key is to find the exact point of controlled convergence.

(Note for user: A schematic diagram plotting "Support Pressure" on the Y-axis against "Radial Displacement" on the X-axis should be drawn here. Show a downward curving line for the Ground Reaction. Draw a steep straight line intercepting the curve early for "Stiff Support installed Early", and a shallower line intercepting the curve later for "Flexible Support installed correctly".)

Q.5. Describe the storage and design principles of rock caverns for underground storage of crude oil and LNG... (10 Marks)

1. Crude Oil Storage (Unlined Rock Caverns)

- **Storage Principle:** This utilizes the **Hydraulic Containment Principle**. Crude oil is lighter than water and insoluble in it. The cavern is excavated deep below the permanent ground water table. As long as the hydrostatic water pressure pushing inward is greater than the vapor/liquid pressure of the oil pushing outward, the groundwater perfectly seals the oil inside the bare rock.
- **Design Principle:** To ensure the rock remains saturated, a "Water Curtain System" (a gallery of horizontal boreholes) is installed above the cavern roof and constantly pressurized. Caverns for heavy crude are designed with a **"D-shaped" cross-section** featuring a sloping floor leading to a sump pit. This geometry concentrates heavy sludge and allows hot water circulation/heating systems to maintain the crude's pumpability.

2. LNG Storage (Refrigerated Caverns)

- **Storage Principle:** LNG/LPG must be stored as a liquid by reducing the temperature drastically (e.g., -162°C for LNG).

- Containment & Design Principle:** The extreme cold freezes the surrounding groundwater, creating a thick "ice ring" in the rock mass fracture system that physically blocks gas migration. Because the rock mass cannot withstand extreme thermal shock without fracturing (which creates convective heat loss), a **Membrane Containment System** is utilized. Insulating panels (like PU foam) are anchored to the concrete-lined cavern walls, covered by a fully sealed, corrugated stainless steel membrane. Furthermore, an extensive drainage system is required during the initial "cool-down" phase to stop water from bleeding inside and forming destructive internal ice blocks.

Q.6. a) Compute Q... (6 Marks)

Given: $J_n = 12$, $J_r = 1$, $J_a = 2$, $J_w = 0.66$, $\text{SRF} = 5$, $J_v = 33$. **1. Compute Q:**
 First, find RQD: $\text{RQD} = 115 - 3.3(J_v) = 115 - 3.3(33) = 115 - 108.9 = 6.1\%$
(Note: In strict application of Barton's Q-system, an RQD < 10 is customarily set to a nominal value of 10. We will calculate using 10). $Q = \left(\frac{10}{12}\right) \times \left(\frac{1}{2}\right) \times \left(\frac{0.66}{5}\right) = 0.833 \times 0.5 \times 0.132 = 0.055$

i) **How the Q value would change in case of tunnel portals?** At tunnel portals, the rock cover is shallow, the rock is typically heavily weathered (J_a increases), and ground stresses are relaxed or highly unfavorable, leading to a massive spike in the Stress Reduction Factor (SRF). This combination drastically **decreases the Q value**, signifying exceptionally poor ground.

ii) **Ground condition and support recommendations:**

- Ground Condition:** A Q-value of 0.055 corresponds to "**Extremely Poor**" to "**Exceptionally Poor**" rock.
- Support Recommendation:** For an 8 m diameter civil tunnel ($\text{ESR} = 1.0$), the equivalent dimension (D_e) is 8. Plotting $D_e = 8$ and $Q = 0.055$ on the Barton support chart falls squarely into the heaviest reinforcement category. The recommendation is **systematic heavy rock bolting paired with thick fiber-reinforced shotcrete (150+ mm)** or potentially even full **cast concrete lining** and steel arches.

Q.6. b) A rock mass has the following characteristics... (4 Marks)

Given: $\sigma_c = 160 \text{ MPa}$, $Q = 25$, $RMR = 35\%$. (Note: $Q = 25$ designates "Good Rock", whereas $RMR = 35\%$ designates "Poor Rock". The empirical formulas will therefore yield substantially different modulus values. Both estimations are shown below).

1. Estimation using RMR System: For rocks with an $RMR < 50$, the Serafim and Pereira (1983) equation is standard: $E_m = 10^{(RMR-10)/40}$ $E_m = 10^{(35-10)/40} = 10^{25/40} = 10^{0.625} \approx 4.22 \text{ GPa}$.

2. Estimation using Q System: Using Barton's correlation incorporating the unconfined compressive strength (σ_c): $E_m = 10 \left(\frac{Q \times \sigma_c}{100} \right)^{1/3}$ $E_m = 10 \left(\frac{25 \times 160}{100} \right)^{1/3} = 10 \times (40)^{1/3} = 10 \times 3.42 \approx 34.20 \text{ GPa}$.